

Low-Cost Biosensor for Disease Detection

1. Introduction

Early detection of diseases is important for effective treatment and prevention. However, many diagnostic methods are expensive and require laboratory facilities. Biosensors are analytical devices that can detect biological substances quickly and at low cost. This project focuses on developing a simple and affordable biosensor for disease detection.

2. Problem Statement

In many rural and low-resource areas, access to advanced diagnostic tools is limited. There is a need for a low-cost, portable, and easy-to-use device that can detect diseases quickly without requiring complex equipment.

3. Objectives

- To design a low-cost biosensor
 - To enable rapid detection of disease biomarkers
 - To make diagnosis accessible in remote areas
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4. Methodology

The proposed biosensor works based on biological recognition and signal detection:

- Select a specific biomarker (DNA/RNA or protein)
- Use a detection system (such as CRISPR-based or enzyme-based method)
- Convert the biological signal into a visible output (color change)
- Use a paper-based strip or simple device for testing

This method allows easy detection without advanced instruments.

5. Expected Outcome

- Fast detection of diseases within minutes
- Affordable diagnostic solution
- Easy-to-use device for non-specialists

6. Applications

- Detection of infectious diseases
- Early diagnosis of health conditions
- Use in rural healthcare and clinics

7. Conclusion

The development of a low-cost biosensor can improve healthcare accessibility by providing quick and affordable disease detection. It has the potential to reduce dependency on expensive laboratory tests and improve early diagnosis.